# PurityLoop Model Results

Slide-ready notes generated from the July 30 model handoff package. Use these for deep-learning result slides and panel explanation.

## Slide 1 - Model Result

# 59.5 mAP@0.5

PurityLoop reached 59.5 mAP@0.5 on 8,453 held-out validation images across 9 waste classes.

Model: remask200_40ep, YOLOv8m-seg, 9-class waste instance segmentation.

Task: Detect individual waste objects and classify their material type, fully in browser via ONNX.

| Metric | Value | Meaning |
|---|---|---|
| mAP@0.5 | 0.595 | Main detection-quality score |
| Precision | 0.606 | Of boxes shown, how many are correct |
| Recall | 0.579 | Of real objects, how many are found |
| Naming accuracy | 0.918 | When detected, class name is usually correct |

Safe explanation: When PurityLoop detects an object, it names the material correctly about 92% of the time, but it detects about 58% of objects present. Use wording: Detected in this photo. Avoid: This bin contains.

## Slide 2 - Classes

PurityLoop detects 9 fixed waste classes:

- plastic
- paper
- cardboard
- metal
- glass
- textile
- food waste
- battery
- general trash

general_trash should be routed to needs review, not shown as a confident label.

## Slide 3 - Why This Model Shipped

Training story: strongest improvement came from fixing validation labels, not making the model bigger.

| Step | Result |
| --- | --- |
| Validation labels audited | +21.9% under-labelled objects found |
| Score before corrected comparison | 0.552 mAP@0.5 |
| Score after correction | 0.595 mAP@0.5 |
| Improvement | +0.043 mAP@0.5 |

Slide wording: Correcting the answer sheet revealed that the model was being penalized for detecting real objects that were missing from labels. After correction, the shipped model improved from 0.552 to 0.595 mAP@0.5.

## Slide 4 - Model Comparison

Use this table to give context. Do not compare 59.5 with COCO or unrelated metric scores.

| Model / System | mAP@0.5 |
| --- | --- |
| ZeroWaste published baseline | 33.5-36.3 |
| TrashDet on TACO | 19.5 |
| Zabble commercial benchmark | 54.5 |
| PurityLoop | 59.5 |

## Slide 5 - Reliability And Generalization

- 0 train/validation collisions after cleanup.
- 8,453 held-out validation images, much larger than many benchmark test splits.
- Training from 40 to 70 epochs improved only +0.004, so longer training was not a useful lever.
- Higher resolutions were tested and rejected; 640 px was faster and performed better for this browser model.

## Slide 6 - Honest Limits

Put limitations on slide. It makes the result more credible.

| Case | Result |
|---|---|
| One close-up object | ~85% found |
| 2-5 objects | ~56% found |
| 6+ cluttered objects | ~42% found |

Small objects and cluttered scenes are the hardest cases.

## Slide 7 - Browser ONNX Proof

- best.onnx
- 54.6 MB fp16
- runs client-side in browser
- no server or GPU needed for inference
- PyTorch-to-ONNX parity checked

Parity result: 312 PyTorch detections vs 312 ONNX detections, with 0/30 images differing in detection count.

## Slide 8 - Human Review / Abstain

PurityLoop should not force a confident answer when the model is unsure.

Route to needs review when:

- top score is too low
- metal and plastic are close
- class is general_trash
- object appears outside the 9 trained classes

Slide wording: In recycling, a confident wrong sort can contaminate a batch. PurityLoop routes uncertain cases to human review instead of guessing.

## Single-Slide Version

- Big headline: 59.5 mAP@0.5
- Subline: 9 classes, 8,453 held-out validation images, browser ONNX
- Metrics: Precision 0.606, Recall 0.579, Naming accuracy 0.918
- Honest limit: Best on one close-up object (~85% found), weaker on cluttered bins (~42% found with 6+ objects)
- Safety wording: Detected in this photo, not This bin contains